
Samsung PIM Simulator Summary

***Intelligent System
Laboratory***

Samsung PIM Simulator CMDs

❑ PIM Instruction Set Architecture

- Simulator의 CMDs

Type	Command	Description	Result (DST)	Operand (SRC0)	Operand (SRC1)
Arithmetic	ADD	Addition	GRF	GRF, BANK, SRF	GRF, BANK, SRF
Arithmetic	MUL	Multiplication	GRF	GRF, BANK	GRF, BANK, SRF
Arithmetic	MAC	Multiply-Accumulate	GRF_B	GRF, BANK	GRF, BANK, SRF
Arithmetic	MAD	Multiply-Add	GRF	GRF, BANK	GRF, BANK, SRF
Data	MOV	load/store data from register to bank	GRF	GRF, SRF	GRF, BANK
Data	FILL	copy data from bank to register	GRF	GRF, BANK	GRF, BANK
Control	NOP	do nothing	-	-	-
Control	JUMP	jump instruction	-	-	-
Control	EXIT	exit instruction	-	-	-

Samsung PIM Simulator CMDs

❑ PIM Instruction Set Architecture

- RISC-style 32비트 Instructions를 Support
- 3가지 Instruction 유형:
 - 4개의 Arithmetic: ADD, MUL, MAC, MAD
 - 2개의 Data Transfer: MOV, FILL
 - 3개의 Control: NOP, JUMP, EXIT
- JUMP Instruction: 0-cycle static branch
 - 미리 programming된 반복 횟수만 지원
- Operand 유형:
 - Vector Register(GRF_A, GRF_B)
 - Scalar Register(SRF)
 - Bank Row Buffer

Samsung PIM Simulator CMDs

❑ PIM Instruction Set Architecture

- PIM Instructions는 CRF에 저장됨, Memory Instructions가 CRF를 trigger하여 target instruction를 수행
 - 각 Memory Instruction이 CRF PC를 increment
- DRAM Instruction이 PIM 연산을 위해 DRAM에서 data를 가져오는 위치를 결정

Samsung PIM Simulator CMDs

❑ Movement of Data

- SB mode: standard DRAM operation
- HAB mode: Allowing concurrent accesses to multiple banks with a single DRAM command
- PIM mode: Triggers the execution of PIM instructions on the CRF by DRAM Command

Mode	Transaction	PIM Instruction	Operation
SB	Read	-	Normal Memory Read
SB	Write	-	Normal Memory Write
HAB	Write	-	PIM Write (Host to PIM Register)
PIM	-	MOV	read or write from bank to PIM Register

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – TEST_F()

- Samsung PIM Simulator는 GoogleTest(gtest)기반의 simulation test를 수행함
- 따라서, flow를 Samsung PIM Simulator가 hbm read operation을 어떻게 test하는지 따라가며 알아본다

./sim --gtest_filter=MemBandwidthFixture.hbm read bandwidth

```
1. #include <bitset>
2. #include <cstdint>
3. #include <iostream>
4. #include <limits>
5. #include <memory>
6. #include <random>
7.
8. #include "Burst.h"
9. #include "MultiChannelMemorySystem.h"
10. #include "gtest/gtest.h"
11. #include "tests/PIMKernel.h"
12. #include "tests/TestCases.h"
13.
14. using namespace std;
15.
16. int main(int argc, char* argv[])
17. {
18.     if (argc > 1)
19.     {
20.         ::testing::InitGoogleTest(&argc, argv);
21.     }
22.     else
23.     {
24.         ::testing::InitGoogleTest(&argc, argv);
25.         testing::GTEST_FLAG(filter) = "-*bw*.*";
26.     }
27.
28.     return RUN_ALL_TESTS();
29. }
```

MainTest.cpp

```
1. TEST_F(MemBandwidthFixture, hbm_read_bandwidth)
2. {
3.     setDataSize(128 * 1024 * 64); // in bytes
4.     uint64_t bw = measureCycle(false);
5.     float effective_bw_ratio = 0.8;
6.     EXPECT_TRUE(bw > 256 * effective_bw_ratio);
7. }
```

MemTestCases.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – SetUp()

- ./sim --gtest_filter=MemBandwidthFixture.hbm_read_bandwidth를 실행 시,
- TEST_F(MemBandwidthFixture, hbm_read_bandwidth)가 실행됨
- 이 때, TEST_F()는 MemBandwidthFixture의 SetUp()을 실행하여 mem에 대한 정보를 initialize함

MemTestCases.cpp

```
1. TEST_F(MemBandwidthFixture, hbm_read_bandwidth)
2. {
3.     setDataSize(128 * 1024 * 64); // in bytes
4.     uint64_t bw = measureCycle(false);
5.     float effective_bw_ratio = 0.8;
6.     EXPECT_TRUE(bw > 256 * effective_bw_ratio);
7. }
```

TestCases.h

```
1. virtual void SetUp()
2. {
3.     cur cycle = 0;
4.     mem = make_shared<MultiChannelMemorySystem>("ini/HBM2_samsung_2M_16B_x64.ini",
5.         "system hbm.ini", ".", "example_app", 256 * 16);
6.     mem_size = (uint64_t)getConfigParam(UINT, "NUM_CHANS") * getConfigParam(UINT, "NUM_RANKS") *
7.         getConfigParam(UINT, "NUM_BANK_GROUPS") * getConfigParam(UINT, "NUM_BANKS") *
8.         getConfigParam(UINT, "NUM_ROWS") * getConfigParam(UINT, "NUM_COLS");
9.     basic_stride =
10.         (getConfigParam(UINT, "JEDEC_DATA_BUS_BITS") * getConfigParam(UINT, "BL") / 8);
11. }
```

```
1. MultiChannelMemorySystem::MultiChannelMemorySystem(const string& deviceIniFilename_,
2.     const string& systemIniFilename_,
3.     const string& pwd_, const string& traceFilename_,
4.     unsigned megsOfMemory_, string* visFilename_)
5. : megsOfMemory(megsOfMemory_),
6.   deviceIniFilename(deviceIniFilename_),
7.   systemIniFilename(systemIniFilename_),
8.   traceFilename(traceFilename_),
9.   pwd(pwd_),
10.  visFilename(visFilename_),
11.  clockDomainCrossover(new ClockDomain::Callback<MultiChannelMemorySystem, void>{
12.      this, &MultiChannelMemorySystem::actual_update}),
13.  csvOut(new CSVWriter(visDataOut))
14. {
15.     currentClockCycle = 0;
16.     if (visFilename)
17.         printf("CC VISFILENAME=%s\n", visFilename->c_str());
18.     if (!isPowerOfTwo(megsOfMemory))
19.     {
20.         ERROR("Please specify a power of 2 memory size");
21.         abort();
22.     }
23.
24.     ConfigurationDB& configDB = ConfigurationDB::getDB();
25.     configDB.initialize();
26.     configDB.updateFromFile(deviceIniFilename);
27.     configDB.updateFromFile(systemIniFilename);
28.
29.     addrMapping = new AddrMapping();
30.     configuration = new Configuration(*addrMapping);
31.     numFence = new unsigned[configuration->NUM_CHANS]();
32.
33.     for (size_t i = 0; i < configuration->NUM_CHANS; i++)
34.     {
35.         MemorySystem* channel = new MemorySystem(i, megsOfMemory / configuration->NUM_CHANS,
36.             (*csvOut), dramsimLog, *configuration);
37.         channels.push_back(channel);
38.     }
39. }
40. }
```

callback 등록

Clock 초기화

각 .ini의 내용을
configDB에 저장

각 .ini의 내용을 기반해
address bit 정보 가져옴

Channel 개수만큼
MemorySystem 생성

MultiChannelMemorySystem.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – SetUp()

- Memory 생성 시
 - Channel개수만큼 MemorySystem 생성 → 각 MemorySystem 마다 Rank개수만큼 Rank Class 생성

```
1. MultiChannelMemorySystem::MultiChannelMemorySystem(const string& deviceIniFilename_,
2.                                                         const string& systemIniFilename_,
3.                                                         const string& pwd_, const string& traceFilename_,
4.                                                         unsigned megsOfMemory_, string* visFilename_)
5. : megsOfMemory(megsOfMemory_),
6.   deviceIniFilename(deviceIniFilename_),
7.   systemIniFilename(systemIniFilename_),
8.   traceFilename(traceFilename_),
9.   pwd(pwd_),
10.  visFilename(visFilename_),
11.  clockDomainCrossover(new ClockDomain::Callback<MultiChannelMemorySystem, void>(
12.      this, &MultiChannelMemorySystem::actual_update)),
13.  csvOut(new CSVWriter(visDataOut))
14. {
15.     currentClockCycle = 0;
16.     if (visFilename)
17.         printf("CC VISFILENAME=%s\n", visFilename->c_str());
18.     if (!isPowerOfTwo(megsOfMemory))
19.     {
20.         ERROR("Please specify a power of 2 memory size");
21.         abort();
22.     }
23. }
24. ConfigurationDB& configDB = ConfigurationDB::getDB();
25. configDB.initialize();
26. configDB.updateFromFile(deviceIniFilename);
27. configDB.updateFromFile(systemIniFilename);
28.
29. addrMapping = new AddrMapping();
30. configuration = new Configuration(*addrMapping);
31. numFence = new unsigned[configuration->NUM_CHANS]();
32.
33. for (size_t i = 0; i < configuration->NUM_CHANS; i++)
34. {
35.     MemorySystem* channel = new MemorySystem(i, megsOfMemory / configuration->NUM_CHANS,
36.                                               ("csvOut", dramsimLog, "configuration");
37.     channels.push_back(channel);
38. }
39.
40. }
```

Channel 개수만큼
MemorySystem 생성

```
1. MemorySystem::MemorySystem(unsigned id, unsigned int megsOfMemory, CSVWriter& csvOut_,
2.                               ostream& simLog, Configuration& configuration)
3. : dramsimLog(simLog),
4.   ReturnReadData(NULL),
5.   WriteDataDone(NULL),
6.   systemID(id),
7.   csvOut(csvOut_),
8.   numOnTheFlyTransactions(0),
9.   config(configuration)
10. {
11.     currentClockCycle = 0;
12.
13.     DEBUG("===== MemorySystem " << systemID << " =====");
14.     uint64_t bytePerRank = static_cast<uint64_t>{
15.         (getConfigParam(UINT, "NUM_ROWS") * getConfigParam(UINT, "DEVICE_WIDTH")) *
16.         (getConfigParam(UINT, "NUM_COLS") * getConfigParam(UINT, "NUM_BANKS")) *
17.         (getConfigParam(UINT64, "JEDEC_DATA_BUS_BITS")) / getConfigParam(UINT, "DEVICE_WIDTH")
18.     };
19.     uint64_t megsOfStoragePerRank = bytePerRank >> 20; // bytes to MB (2^20 = 1,048,576 = Mega)
20.
21.     num_ranks_ = (megsOfMemory / Byte2MB(bytePerRank));
22.
23.     // If this is set, effectively override the number of ranks
24.     if (megsOfMemory != 0)
25.     {
26.         num_ranks_ = (num_ranks_ > 0) ? num_ranks_ : 1;
27.         if (num_ranks_ == 0)
28.         {
29.             PRINT("WARNING: Cannot create memory system with "
30.                 << megsOfMemory << "MB, defaulting to minimum size of " << megsOfStoragePerRank
31.                 << "MB");
32.         }
33.     }
34.     setDevConfigParam(UINT, "NUM_RANKS", num_ranks_);
35.     // FIXME, need to synchronize between ConfigurationDB and Configuration
36.     config.NUM_RANKS = num_ranks_;
37. }
38. }
```

```
39. unsigned num_devices =
40.     (getConfigParam(UINT, "JEDEC_DATA_BUS_BITS") / getConfigParam(UINT, "DEVICE_WIDTH"));
41. setDevConfigParam(UINT, "NUM_DEVICES", num_devices);
42. setDevConfigParam(UINT64, "TOTAL_STORAGE", num_ranks_ * Byte2MB(bytePerRank));
43.
44. PRINT("CH. " << systemID << " TOTAL_STORAGE : " << getConfigParam(UINT64, "TOTAL_STORAGE")
45.       << "MB | " << getConfigParam(UINT, "NUM_RANKS") << " Ranks | "
46.       << getConfigParam(UINT, "NUM_DEVICES") << " Devices per rank");
47.
48. memoryController = new MemoryController(this, csvOut, dramsimLog, config);
49.
50. // TODO: change to other vector constructor?
51. ranks = new vector<Rank*>();
52.
53. for (size_t i = 0; i < num_ranks_; i++)
54. {
55.     Rank* r = new Rank(dramsimLog, config);
56.     r->setChanId(systemID);
57.     r->setRankId(i);
58.     r->attachMemoryController(memoryController);
59.     r->pimRank->setChanId(systemID);
60.     r->pimRank->setRankId(i);
61.     ranks->push_back(r);
62. }
63.
64. memoryController->attachRanks(ranks);
65. }
```

MultiChannelMemorySystem.cpp

MemorySystem.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – SetUp()

- Memory 생성 시

- Rank개수만큼 Rank Class 생성 → Rank는 자신의 Class 변수에 channel, rank id 및 Bank들 관련 변수 초기화

```
1. MemorySystem::MemorySystem(unsigned id, unsigned int megsOfMemory, CSVWriter& csvOut,
2.   ostream& simLog, Configuration& configuration)
3. : dramsimLog(simLog),
4.   ReturnReadData(NULL),
5.   WriteDataDone(NULL),
6.   systemID(id),
7.   csvOut(csvOut),
8.   numOnTheFlyTransactions(0),
9.   config(configuration)
10. {
11.   currentClockCycle = 0;
12.
13.   DEBUG("==== MemorySystem " << systemID << " =====");
14.   uint64_t bytePerRank = static_cast<uint64_t>{
15.     (getConfigParam(UINT64, "NUM_ROWS") *
16.      getConfigParam(UINT, "NUM_COLS") * getConfigParam(UINT, "DEVICE_WIDTH")) *
17.     getConfigParam(UINT, "NUM_BANKS") *
18.     (getConfigParam(UINT64, "JEDEC_DATA_BUS_BITS") / getConfigParam(UINT, "DEVICE_WIDTH")) /
19.     8);
20.   uint64_t megsOfStoragePerRank = bytePerRank >> 20; // bytes to MB (2^20 = 1,048,576 = Mega)
21.
22.   num_ranks_ = (megsOfMemory / Byte2MB(bytePerRank));
23.
24.   // If this is set, effectively override the number of ranks
25.   if (megsOfMemory != 0)
26.   {
27.     num_ranks_ = (num_ranks_ > 0) ? num_ranks_ : 1;
28.     if (num_ranks_ == 0)
29.     {
30.       PRINT("WARNING: Cannot create memory system with "
31.         << megsOfMemory << "MB, defaulting to minimum size of " << megsOfStoragePerRank
32.         << "MB");
33.     }
34.     setDevConfigParam(UINT, "NUM_RANKS", num_ranks_);
35.     // FIXME, need to synchronize between ConfigurationDB and Configuration
36.     config.NUM_RANKS = num_ranks_;
37.   }
38. }
```

MemorySystem.cpp

```
39. unsigned num_devices =
40.   (getConfigParam(UINT, "JEDEC_DATA_BUS_BITS") / getConfigParam(UINT, "DEVICE_WIDTH"));
41. setDevConfigParam(UINT, "NUM_DEVICES", num_devices);
42. setDevConfigParam(UINT64, "TOTAL_STORAGE", num_ranks_ * Byte2MB(bytePerRank));
43.
44. PRINT("CH. " << systemID << " TOTAL_STORAGE : " << getConfigParam(UINT64, "TOTAL_STORAGE")
45.   << "MB | " << getConfigParam(UINT, "NUM_RANKS") << " Ranks | "
46.   << getConfigParam(UINT, "NUM_DEVICES") << " Devices per rank");
47.
48. memoryController = new MemoryController(this, csvOut, dramsimLog, config);
49.
50. // TODO: change to other vector constructor?
51. ranks = new vector<Rank*>();
52.
53. for (size_t i = 0; i < num_ranks_; i++)
54. {
55.   Rank* r = new Rank(dramsLog, config);
56.   r->setChanId(systemID);
57.   r->setRankId(i);
58.   r->attachMemoryController(memoryController);
59.   r->pimRank->setChanId(systemID);
60.   r->pimRank->setRankId(i);
61.   ranks->push_back(r);
62. }
63.
64. memoryController->attachRanks(ranks);
65. }
```

```
1. Rank::Rank(ostream& simLog, Configuration& configuration)
2. : chanId(-1),
3.   rankId(-1),
4.   dramsimLog(simLog),
5.   isPowerDown(false),
6.   refreshWaiting(false),
7.   readReturnCountdown(0),
8.   banks(getConfigParam(UINT, "NUM_BANKS"), Bank(simLog)),
9.   bankStates(getConfigParam(UINT, "NUM_BANKS"), BankState(simLog)),
10.  config(configuration),
11.  outgoingDataPacket(NULL),
12.  dataCyclesLeft(0),
13.  mode_(dramMode::SB)
14. {
15. {
16.   memoryController = NULL;
17.   currentClockCycle = 0;
18.   abmr1Even_ = abmr1Odd_ = abmr2Even_ = abmr2Odd_ = sbmr1_ = sbmr2_ = false;
19.
20.   pimRank = make_shared<PIMRank>(dramsLog, config);
21.   pimRank->attachRank(this);
22. }
```

Rank.cpp

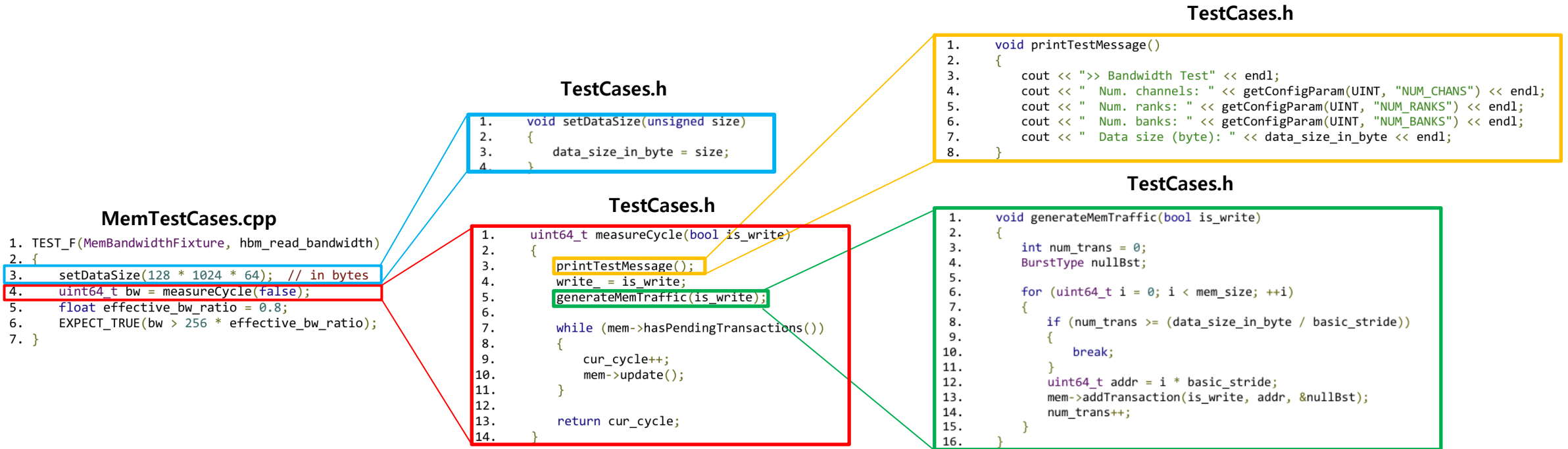
```
1. class Rank : public SimulatorObject
2. {
3. private:
4.   int chanId;
5.   int rankId;
6.   ostream& dramsimLog;
7.   bool isPowerDown;
8.   Configuration& config;
9.
10. public:
11.   // functions
12.   Rank(ostream& simLog, Configuration& configuration);
13.   virtual ~Rank();
14.
15.   void receiveFromBus(BusPacket* packet);
16.   void check(BusPacket* packet);
17.   void updateState(BusPacket* packet);
18.   void sendToBank(BusPacket* packet);
19.
20.   void checkBank(BusPacketType type, int bank, int row);
21.   void updateBank(BusPacketType type, int bank, int row, bool targetBank, bool targetBankgroup);
22.   void attachMemoryController(MemoryController* mc);
23.   int getChanId() const;
24.   void setChanId(int id);
25.   int getRankId() const;
26.   void setRankId(int id);
27.   void update();
28.   void powerUp();
29.   void powerDown();
30.
31.   void readSb(BusPacket* packet);
32.   void writeSb(BusPacket* packet);
33.
34. // fields
35. MemoryController* memoryController;
36. BusPacket* outgoingDataPacket;
37. shared_ptr<PIMRank> pimRank;
38. unsigned dataCyclesLeft;
39. bool refreshWaiting;
40.
41. // these are vectors so that each element is per-bank
42. vector<BusPacket*> readReturnPacket;
43. vector<unsigned> readReturnCountdown;
44.
45. vector<Bank> banks;
46. vector<BankState> bankStates;
47.
48. dramMode mode_;
49. bool abmr1Even_, abmr1Odd_, abmr2Even_, abmr2Odd_, sbmr1_, sbmr2_;
50.
51. const char* getModeColor()
52. {
53.   switch (mode_)
54.   {
55.     case dramMode::SB:
56.       return END;
57.     case dramMode::HAB:
58.       return GREEN;
59.     case dramMode::HAB_PIM:
60.       return CYAN;
61.   }
62.   return GRAY;
63. }
64. ;;
```

Rank.h

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- mem 초기화 이후 data_size_in_byte를 설정
- **measureCycle()**을 통하여 성능을 simulate
- **generateMemTraffic()**은 memory transaction을 생성하는 함수 → Memory Trace를 생성하는 역할과 동일



Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle() → generateMemTraffic()

- **generateMemTraffic()**은 어떤, 얼마만큼의 memory transaction(Read or Write)을 수행할 것인지 결정하여 transactionQueue를 채움
- transactionQueue는 최종적으로 MemoryController.cpp에서 관리하며, scheduling logic에 따라 issue됨
- **MultiChannelMemorySystem::findChannelNumber()**은 다음 page에서 설명
- mem_size 및 basic_stride는 PPT page7의 SetUp()에서 정의됨
 - $\text{mem_size} = \text{NUM_CHANS} \times \text{NUM_RANKS} \times \text{NUM_BANK_GROUPS} \times \text{NUM_BANKS} \times \text{NUM_ROWS} \times \text{NUM_COLS}$
 - $\text{basic_stride} = (\text{JEDEC_DATA_BUS_BITS} \times \text{BL}) / 8$ [ex. JEDEC_DATA_BUS_BITS = 64, BL = 4, basic_stride = (64 bits × 4) / 8 bytes/bit = 32 bytes (= 한 번의 READ/WRITE transaction 시 전송되는 data 크기)]

TestCases.h

```
1. void generateMemTraffic(bool is_write)
2. {
3.     int num_trans = 0;
4.     BurstType nullBst;
5.
6.     for (uint64_t i = 0; i < mem_size; ++i)
7.     {
8.         if (num_trans >= (data_size_in_byte / basic_stride))
9.         {
10.            break;
11.        }
12.        uint64_t addr = i * basic_stride;
13.        mem->addTransaction(is_write, addr, &nullBst);
14.        num_trans++;
15.    }
16. }
```

MultiChannelMemorySystem.cpp

```
1. bool MultiChannelMemorySystem::addTransaction(bool isWrite, uint64_t addr, BurstType* data)
2. {
3.     unsigned channelNumber = findChannelNumber(addr);
4.     return channels[channelNumber]->addTransaction(isWrite, addr, data);
5. }
```

```
1. unsigned MultiChannelMemorySystem::findChannelNumber(uint64_t addr)
2. {
3.     // Single channel case is a trivial shortcut case
4.     if (configuration->NUM_CHANS == 1)
5.     {
6.         return 0;
7.     }
8.
9.     if (!isPowerOfTwo(configuration->NUM_CHANS))
10.    {
11.        ERROR("We can only support power of two # of channels.\n"
12.            << "I don't know what Intel was thinking, but trying to address map half"
13.            << " a bit is a neat trick that we're not sure how to do");
14.        abort();
15.    }
16.
17.    // only chan is used from this set
18.    unsigned channelNumber, rank, bank, row, col;
19.    addrMapping->addressMapping(addr, channelNumber, rank, bank, row, col);
20.    if (channelNumber >= configuration->NUM_CHANS)
21.    {
22.        ERROR("Got channel index " << channelNumber << " but only " << configuration->NUM_CHANS
23.            << " exist");
24.        abort();
25.    }
26.    return channelNumber;
27. }
```

MultiChannelMemorySystem.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle() → generateMemTraffic()

- **addressMapping**의 경우 주어진 address를 shifting 및 slicing함으로써 처리됨
- **addressMapping**에서 address의 하위 6bit(= transaction size 및 burst size 나타내는 bit)는 shift right하여 제거해야 함
 - Burst: 한 번에 읽는 작은 데이터 덩어리 (DDR에서 보통 8byte), Transaction: 전체 요청 단위, 여러 burst로 구성, 보통 cache line 크기와 동일(= 64byte)
 - Controller는 Transaction 1개를 받아서 처리 & Column Address는 Burst 중 Byte마다 +1 증가 → 따라서, 64byte = 2^6 byte → 6 bit right shift 필요
- Shift하고 남은 bit들은 적절히 slicing하여 scheme에 따라, (channel : rank : bank : row : col)로 분리 (scheme마다 slice방법 다름)
- **MultiChannelMemorySystem::findChannelNumber()**는 channelNumber(= channel id)를 반환

MultiChannelMemorySystem.cpp

```
1. unsigned MultiChannelMemorySystem::findChannelNumber(uint64_t addr)
2. {
3.     // Single channel case is a trivial shortcut case
4.     if (configuration->NUM_CHANS == 1)
5.     {
6.         return 0;
7.     }
8.
9.     if (!isPowerOfTwo(configuration->NUM_CHANS))
10.    {
11.        ERROR("We can only support power of two # of channels.\n")
12.        << "I don't know what Intel was thinking, but trying to address map half"
13.        << " a bit is a neat trick that we're not sure how to do";
14.        abort();
15.    }
16.
17.    // only chan is used from this set
18.    unsigned channelNumber, rank, bank, row, col;
19.    addrMapping->addressMapping(addr, channelNumber, rank, bank, row, col);
20.    if (channelNumber >= configuration->NUM_CHANS)
21.    {
22.        ERROR("Got channel index " << channelNumber << " but only " << configuration->NUM_CHANS
23.        << " exist");
24.        abort();
25.    }
26.    return channelNumber;
27. }
```

addressMapping.cpp

```
1. void AddrMapping::addressMapping(uint64_t physicalAddress, unsigned& newTransactionChan,
2. unsigned& newTransactionRank, unsigned& newTransactionBank,
3. unsigned& newTransactionRow, unsigned& newTransactionColumn)
4. {
5.     if ((physicalAddress & transactionMask) != 0)
6.     {
7.         DEBUG("WARNING: address 0x" << std::hex << physicalAddress << std::dec
8.         << " is not aligned to the request size of "
9.         << transactionSize);
10.    }
11.    // each burst will contain JEDEC_DATA_BUS_BITS/8 bytes of data, so the bottom
12.    // bits (3 bits for a single channel DDR system) are thrown away before mapping the other bits
13.    physicalAddress >>= byteOffsetWidth;
14.
15.    // The next thing we have to consider is that when a request is made for a
16.    // we've taken into account the granularity of a single burst by shifting
17.    // off the bottom 3 bits, but a transaction has to take into account the
18.    // burst length (i.e. the requests will be aligned to cache line sizes which
19.    // should be equal to transactionSize above).
20.    // Since the column address increments internally on bursts, the bottom n
21.    // bits of the column (colLow) have to be zero in order to account for the
22.    // total size of the transaction. These n bits should be shifted off the
23.    // address and also subtracted from the total column width.
24.    // zero. These zero bits must be made up of the byte offset (3 bits) and
25.    // also
26.    // from the bottom bits of the column
27.    // For example: colLowBits = log2(64bytes) - 3 bits = 3 bits
28.    physicalAddress >>= colLowBitWidth;
29.    if (DEBUG_ADDR_MAP)
30.    {
31.        DEBUG("Bit widths: ch:" << channelBitWidth << " r:" << rankBitWidth << " b:" << bankBitWidth
32.        << " row:" << rowBitWidth << " colLow:" << colLowBitWidth
33.        << " colHigh:" << colHighBitWidth << " off:" << byteOffsetWidth
34.        << " Total:"
35.        << "(channelBitWidth + rankBitWidth + bankBitWidth + rowBitWidth +
36.        colLowBitWidth + colHighBitWidth + byteOffsetWidth)");
37.    }
38.
39.    // perform various address mapping schemes
40.    if (addressMappingScheme == Scheme1)
41.    {
42.        // chan:rank:row:col:bank
43.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
44.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
45.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
46.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
47.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
48.    }
49.    else if (addressMappingScheme == Scheme2)
50.    {
51.        // chan:row:col:bank:rank
52.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
53.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
54.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
55.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
56.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
57.    }
58.    else if (addressMappingScheme == Scheme3)
59.    {
60.        // chan:rank:bank:col:row
61.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
62.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
63.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
64.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
65.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
66.    }
67.    else if (addressMappingScheme == Scheme4)
68.    {
69.        // chan:rank:bank:row:col
70.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
71.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
72.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
73.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
74.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
75.    }
76.    else if (addressMappingScheme == Scheme5)
77.    {
78.        // chan:row:col:rank:bank
79.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
80.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
81.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
82.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
83.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
84.    }
85.    else if (addressMappingScheme == Scheme6)
86.    {
87.        // chan:row:bank:rank:col
88.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
89.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
90.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
91.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
92.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
93.    }
94.    else if (addressMappingScheme == Scheme7)
95.    {
96.        // chan:row:bank:rank:col
97.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
98.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
99.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
100.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
101.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
102.    }
103.    else if (addressMappingScheme == Scheme8)
104.    {
105.        // row:col:rank:bank:chan
106.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
107.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
108.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
109.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
110.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
111.    }
112.    else if (addressMappingScheme == Scheme9)
113.    {
114.        // row:col:rank:bank:chan
115.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
116.        newTransactionBank = diffBitWidth(physicalAddress, bankBitWidth);
117.        newTransactionRank = diffBitWidth(physicalAddress, rankBitWidth);
118.        newTransactionColumn = diffBitWidth(physicalAddress, colHighBitWidth);
119.        newTransactionRow = diffBitWidth(physicalAddress, rowBitWidth);
120.        newTransactionChan = diffBitWidth(physicalAddress, channelBitWidth);
121.    }
122.    else
123.    {
124.        ERROR("Error - Unknown Address Mapping Scheme");
125.        exit(-1);
126.    }
127.
128.    if (DEBUG_ADDR_MAP)
129.    {
130.        DEBUG("Mapped Ch=" << newTransactionChan << " Rank=" << newTransactionRank
131.        << " Bank=" << newTransactionBank << " Row=" << newTransactionRow
132.        << " Col=" << newTransactionColumn << "\n");
133.    }
134. }
```

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle() → generateMemTraffic()

- **addTransaction()**은 해당 transaction의 addr가 나타내는 **channel의 Memory Controller**에게 해당 transaction을 넘김
- Memory Controller는 **transactionQueue**가 **꽉 차지 않았으면**, 해당 transaction을 Queue에 저장
- 꽉 찼으면 pendingTransactions 배열에 저장됨
- **trans->timeAdded**는 기존 DRAMSim2에서 latency 지표 계산에 쓰임
 - But, DRAMSim2에서만 **trans->timeAdded**가 statistic에 반영되고, PIMSimulator는 이를 반영하지 않음 (사실상 안 쓰이는 변수)

MultiChannelMemorySystem.cpp

```
1. bool MultiChannelMemorySystem::addTransaction(bool isWrite, uint64_t addr, BurstType* data)
2. {
3.     unsigned channelNumber = findChannelNumber(addr);
4.     return channels[channelNumber]->addTransaction(isWrite, addr, data);
5. }
```

MemorySystem.cpp

```
1. bool MemorySystem::addTransaction(bool isWrite, uint64_t addr, BurstType* data)
2. {
3.     TransactionType type = isWrite ? DATA_WRITE : DATA_READ;
4.     Transaction* trans = new Transaction(type, addr, data);
5.     return addTransaction(trans);
6. }
```

MemorySystem.cpp

```
1. bool MemorySystem::addTransaction(Transaction* trans)
2. {
3.     if (memoryController->WillAcceptTransaction())
4.     {
5.         return memoryController->addTransaction(trans);
6.     }
7.     else
8.     {
9.         pendingTransactions.push_back(trans);
10.        return true;
11.    }
12. }
```

MemoryController.cpp

```
1. bool MemoryController::WillAcceptTransaction()
2. {
3.     return transactionQueue.size() < getConfigParam(UINT, "TRANS_QUEUE_DEPTH");
4. }
```

MemoryController.cpp

```
1. bool MemoryController::addTransaction(Transaction* trans)
2. {
3.     if (WillAcceptTransaction())
4.     {
5.         parentMemorySystem->numOnTheFlyTransactions++;
6.         trans->timeAdded = currentClockCycle;
7.         transactionQueue.push_back(trans);
8.         return true;
9.     }
10.    else
11.    {
12.        return false;
13.    }
14. }
```


Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle() → update()

- **measureCycle()**을 통하여 성능을 simulate
- CPU clock과 DRAM clock의 **clock domain**을 정렬하며, **callback**을 호출
- 등록된 **callback**은 **actual_update()** 임 (PPT page7 참고)

TestCases.h

```
1. uint64_t measureCycle(bool is_write)
2. {
3.     printTestMessage();
4.     write_ = is_write;
5.     generateMemTraffic(is_write);
6.
7.     while (mem->hasPendingTransactions())
8.     {
9.         cur_cycle++;
10.        mem->update();
11.    }
12.
13.    return cur_cycle;
14. }
```

```
1. void MultiChannelMemorySystem::update()
2. {
3.     clockDomainCrossover.update();
4. }
5.
6. void MultiChannelMemorySystem::actual_update()
7. {
8.     if (currentClockCycle == 0)
9.     {
10.        InitOutputFiles(traceFilename);
11.        DEBUG("DRAMSim2 Clock Frequency = " << clockDomainCrossover.clock1
12.              << "Hz, CPU Clock Frequency="
13.              << clockDomainCrossover.clock2 << "Hz");
14.    }
15.
16.    if (currentClockCycle > 0 && currentClockCycle % configuration->EPOCH_LENGTH == 0)
17.    {
18.        (*csvOut) << "ms" << currentClockCycle * configuration->tCK * 1E-6;
19.        for (size_t i = 0; i < configuration->NUM_CHANS; i++)
20.        {
21.            channels[i]->printStats(false);
22.        }
23.        csvOut->finalize();
24.    }
25.
26.    for (size_t i = 0; i < configuration->NUM_CHANS; i++)
27.    {
28.        channels[i]->update();
29.    }
30.
31.    currentClockCycle++;
32. }
```

MultiChannelMemorySystem.cpp

```
1. void MultiChannelMemorySystem::setCPUClockSpeed(uint64_t cpuClkFreqHz)
2. {
3.     uint64_t dramsimClkFreqHz = (uint64_t)(1.0 / (configuration->tCK * 1e-9));
4.     clockDomainCrossover.clock1 = dramsimClkFreqHz; // ← DRAM 클럭 (Hz)
5.     clockDomainCrossover.clock2 = cpuClkFreqHz; // ← CPU 클럭 (Hz)
6. }
```

```
1. namespace ClockDomain
2. {
3.     // "Default" crosser with a 1:1 ratio
4.     ClockDomainCrossover::ClockDomainCrossover(ClockUpdateCB* _callback)
5.         : callback(_callback), clock1(1UL), clock2(1UL), counter1(0UL), counter2(0UL)
6. }
```

```
1. void ClockDomainCrossover::update()
2. {
3.     // short circuit case for 1:1 ratios
4.     if (clock1 == clock2 && callback)
5.     {
6.         (*callback)();
7.         return;
8.     }
9.
10.    // Update counter 1.
11.    counter1 += clock1;
12.
13.    while (counter2 < counter1)
14.    {
15.        counter2 += clock2;
16.        // PRINT("CALLBACK: counter1= " << counter1 << "; counter2= " << counter2/ << "; ");
17.        // PRINT("callback address: " << (uint64_t)callback);
18.        if (callback)
19.        {
20.            // PRINT("Callback() " << (uint64_t)callback << "Counters:1=" << counter1 << ",
21.            // 2=" << counter2);
22.            (*callback)();
23.        }
24.    }
25.
26.    if (counter1 == counter2)
27.    {
28.        counter1 = 0;
29.        counter2 = 0;
30.    }
31. }
```

ClockDomain.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → MultiChannelMemorySystem::actual_update() → MemorySystem::update() → Rank::update()
- Rank::update() – 특이사항: 현재 Simulator는 독립적인 command/read/write bus를 사용하고 있음
 1. Read Latency 대기 (ex. config.RL = 20 → 20 cycle 기다린 후 data를 bus에 올림)
 2. Data bus timing 관리 (= BL/2 설정) → Bus에 올라간 data는 BL/2 cycle간 transfer
 3. 실제로 BL/2 cycle만큼 대기 후 완료 시 memoryController에게 완료를 알림

```
6. void MultiChannelMemorySystem::actual_update()
7. {
8.     if (currentClockCycle == 0)
9.     {
10.         InitOutputFiles(traceFilename);
11.         DEBUG("DRAMSim2 Clock Frequency = " << clockDomainCrosser.clock1
12.             << "Hz, CPU Clock Frequency="
13.             << "Hz, CPU Clock Frequency="
14.             << clockDomainCrosser.clock2 << "Hz");
15.     }
16.     if (currentClockCycle > 0 && currentClockCycle % configuration->EPOCH_LENGTH == 0)
17.     {
18.         (*csvOut) << "ms" << currentClockCycle * configuration->tCK * 1E-6;
19.         for (size_t i = 0; i < configuration->NUM_CHANS; i++)
20.         {
21.             channels[i]->printStats(false);
22.         }
23.         csvOut->finalize();
24.     }
25.     for (size_t i = 0; i < configuration->NUM_CHANS; i++)
26.     {
27.         channels[i]->update();
28.     }
29.     currentClockCycle++;
30. }
31.
32. }
```

MultiChannelMemorySystem.cpp

```
1. // update the memory systems state
2. void MemorySystem::update()
3. {
4.     // PRINT(" ----- Memory System Update -----");
5.
6.     // updates the state of each of the objects
7.     // NOTE - do not change order
8.     for (size_t i = 0; i < num_ranks_; i++)
9.     {
10.         (*ranks)[i]->update();
11.     }
12.
13.     // pendingTransactions will only have stuff in it if MARSS is adding stuff
14.     if (pendingTransactions.size() > 0 && memoryController->WillAcceptTransaction())
15.     {
16.         memoryController->addTransaction(pendingTransactions.front());
17.         pendingTransactions.pop_front();
18.     }
19.     memoryController->update();
20.
21.     // simply increments the currentClockCycle field for each object
22.     for (size_t i = 0; i < num_ranks_; i++)
23.     {
24.         (*ranks)[i]->step();
25.     }
26.     memoryController->step();
27.     this->step();
28.
29.     // PRINT("\n"); // two new lines
30. }
```

MemorySystem.cpp

```
1. void Rank::update()
2. {
3.     // An outgoing packet is one that is currently sending on the bus
4.     // do the book keeping for the packet's time left on the bus
5.     if (outgoingDataPacket != NULL)
6.     {
7.         dataCyclesLeft--;
8.         if (dataCyclesLeft == 0)
9.         {
10.            // if the packet is done on the bus, call receiveFromBus and free up
11.            // the bus
12.            memoryController->receiveFromBus(outgoingDataPacket);
13.            outgoingDataPacket = NULL;
14.        }
15.    }
16.
17.    // decrement the counter for all packets waiting to be sent back
18.    for (size_t i = 0; i < readReturnCountdown.size(); i++) readReturnCountdown[i]--;
19.
20.    if (readReturnCountdown.size() > 0 && readReturnCountdown[0] == 0)
21.    {
22.        // RL time has passed since the read was issued; this packet is
23.        // ready to go out on the bus
24.
25.        outgoingDataPacket = readReturnPacket[0];
26.        dataCyclesLeft = config.BL / 2;
27.
28.        // remove the packet from the ranks
29.        readReturnPacket.erase(readReturnPacket.begin());
30.        readReturnCountdown.erase(readReturnCountdown.begin());
31.
32.        if (DEBUG_BUS)
33.        {
34.            PRINT(" -- R" << getChanId() << " Issuing On Data Bus : ");
35.            outgoingDataPacket->print();
36.            PRINT("");
37.        }
38.    }
39.    pimRank->update();
40.    pimRank->step();
41. }
```

Rank.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → MultiChannelMemorySystem::actual_update() → MemorySystem::update() → MemoryController::update()
- Transaction Queue에 미처 올리지 못한 transaction을 올릴 수 있는 지 확인
- 그 후, memoryController::update()를 호출
- 다음 page에서 memoryController::update() 동작을 분석

```
1. // update the memory systems state
2. void MemorySystem::update()
3. {
4.     // PRINT(" ----- Memory System Update -----");
5.
6.     // updates the state of each of the objects
7.     // NOTE - do not change order
8.     for (size_t i = 0; i < num_ranks_; i++)
9.     {
10.        (*ranks)[i] -> update();
11.    }
12.
13.    // pendingTransactions will only have stuff in it if MARSS is adding stuff
14.    if (pendingTransactions.size() > 0 && memoryController->WillAcceptTransaction())
15.    {
16.        memoryController->addTransaction(pendingTransactions.front());
17.        pendingTransactions.pop_front();
18.    }
19.    memoryController->update();
20.
21.    // simply increments the currentClockCycle field for each object
22.    for (size_t i = 0; i < num_ranks_; i++)
23.    {
24.        (*ranks)[i] -> step();
25.    }
26.    memoryController->step();
27.    this->step();
28.
29.    // PRINT("\n"); // two new lines
30. }
```

MemorySystem.cpp

```
1. void MemoryController::update()
2. {
3.     updateBankState();
4.     // check for outgoing command packets and handle countdowns
5.     if (outgoingCmdPacket != NULL)
6.     {
7.         cmdCyclesLeft--;
8.         if (cmdCyclesLeft == 0) // packet is ready to be received by rank
9.         {
10.            (*ranks)[outgoingCmdPacket->rank] -> receiveFromBus(outgoingCmdPacket);
11.            outgoingCmdPacket = NULL;
12.        }
13.    }
14.
15.    // check for outgoing data packets and handle countdowns
16.    if (outgoingDataPacket != NULL)
17.    {
18.        dataCyclesLeft--;
19.        if (dataCyclesLeft == 0)
20.        {
21.            // inform upper levels that a write is done
22.            if (parentMemorySystem->WriteDataDone != NULL)
23.            {
24.                (*parentMemorySystem->WriteDataDone)(parentMemorySystem->systemID,
25.                outgoingDataPacket->physicalAddress,
26.                currentClockCycle);
27.            }
28.            parentMemorySystem->numOnTheFlyTransactions--;
29.            (*ranks)[outgoingDataPacket->rank] -> receiveFromBus(outgoingDataPacket);
30.            outgoingDataPacket = NULL;
31.        }
32.    }
33. }
```

```
34. // if any outstanding write data needs to be sent
35. // and the appropriate amount of time has passed (WL)
36. // then send data on bus
37. //
38. // write data held in fifo vector along with countdowns
39. if (writeDataCountdown.size() > 0)
40. {
41.     for (size_t i = 0; i < writeDataCountdown.size(); i++) writeDataCountdown[i]--;
42.
43.     if (writeDataCountdown[0] == 0)
44.     {
45.         // send to bus and print debug stuff
46.         if (DEBUG_BUS)
47.         {
48.             PRINTN(" -- MC Issuing On Data Bus : ");
49.             writeDataToSend[0] -> print();
50.         }
51.
52.         // queue up the packet to be sent
53.         if (outgoingDataPacket != NULL)
54.         {
55.             ERROR("Error - Data Bus Collision");
56.             exit(-1);
57.         }
58.
59.         outgoingDataPacket = writeDataToSend[0];
60.         dataCyclesLeft = config.BL / 2;
61.
62.         totalTransactions++;
63.
64.         writeDataCountdown.erase(writeDataCountdown.begin());
65.         writeDataToSend.erase(writeDataToSend.begin());
66.     }
67. }
68.
69. // if its time for a refresh issue a refresh
70. // else pop from command queue if it's not empty
71. updateRefresh();
72.
73. // pass a pointer to a poppedBusPacket
74. // function returns true if there is something valid in poppedBusPacket
75. if (commandQueue.pop(&poppedBusPacket))
76. {
77.     updateCommandQueue(poppedBusPacket);
78. }
79. }
```

```
80. updateTransactionQueue();
81.
82. // check for outstanding data to return to the CPU
83. if (returnTransaction.size() > 0)
84. {
85.     if (DEBUG_BUS)
86.     {
87.         PRINTN(" -- MC Issuing to CPU bus : " << *returnTransaction[0]);
88.         totalTransactions++;
89.     }
90.     bool foundMatch = false;
91.     // find the pending read transaction to calculate latency
92.     for (size_t i = 0; i < pendingReadTransactions.size(); i++)
93.     {
94.         if (pendingReadTransactions[i] -> address == returnTransaction[0] -> address)
95.         {
96.             unsigned chan, rank, bank, row, col;
97.             config.addrMapping.addressMapping(returnTransaction[0] -> address, chan, rank, bank,
98.             row, col);
99.             memoryContStats->insertHistogram(
100.             currentClockCycle - pendingReadTransactions[i] -> timeAdded, rank, bank);
101.             // FIXME: Is it correct?
102.             // memcpy(pendingReadTransactions[i] -> data,
103.             // returnTransaction[0] -> data, config.BL * (JEDEC_DATA_BUS_BITS / 8));
104.             returnReadData(pendingReadTransactions[i]);
105.
106.             delete pendingReadTransactions[i];
107.             pendingReadTransactions.erase(pendingReadTransactions.begin() + i);
108.             foundMatch = true;
109.             break;
110.         }
111.     }
112.     if (!foundMatch)
113.     {
114.         ERROR("Can't find a matching transaction for 0x" << hex << returnTransaction[0] -> address
115.         << dec);
116.         abort();
117.     }
118.     delete returnTransaction[0];
119.     returnTransaction.erase(returnTransaction.begin());
120. }
121.
122. // decrement refresh counters
123. for (size_t i = 0; i < config.NUM_RANKS; i++) refreshCountdown[i]--;
124. for (size_t i = 0; i < config.NUM_BANKS; i++) refreshCountdownBank[i]--;
125.
126. // print debug
127. printDebugOnlyUpdate();
128. commandQueue.step();
129. }
```

MemoryController.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → MultiChannelMemorySystem::actual_update() → MemorySystem::update() → MemoryController::update()

1. DRAM Bank들의 state 변화가 완료되었는지 확인

- 모든 Rank와 Bank를 순회
- Count down: 이전 cycle에서 PRECHARGE나 REFRESH 명령이 수행되었다면, 해당 명령의 완료 시간(ex. tRP, tRFC)만큼 counter가 설정되어 있음
- State transition: 매 cycle 1씩 감소하다가 0이 되면, 해당 Bank의 상태를 Idle로 변경하여 다음 명령(ex. ACTIVATE)을 받을 수 있는 상태로 만듦

```
1. void MemoryController::update()
2. {
3.     updateBankState();
4.     // check for outgoing command packets and handle countdowns
5.     if (outgoingCmdPacket != NULL)
6.     {
7.         cmdCyclesLeft--;
8.         if (cmdCyclesLeft == 0) // packet is ready to be received by rank
9.         {
10.            (*ranks)[outgoingCmdPacket->rank]->receiveFromBus(outgoingCmdPacket);
11.            outgoingCmdPacket = NULL;
12.        }
13.    }
14.    // check for outgoing data packets and handle countdowns
15.    if (outgoingDataPacket != NULL)
16.    {
17.        dataCyclesLeft--;
18.        if (dataCyclesLeft == 0)
19.        {
20.            // inform upper levels that a write is done
21.            if (parentMemorySystem->WriteDataDone != NULL)
22.            {
23.                (*parentMemorySystem->WriteDataDone)(parentMemorySystem->systemID,
24.                outgoingDataPacket->physicalAddress,
25.                currentClockCycle);
26.            }
27.            parentMemorySystem->numOnTheFlyTransactions--;
28.            (*ranks)[outgoingDataPacket->rank]->receiveFromBus(outgoingDataPacket);
29.            outgoingDataPacket = NULL;
30.        }
31.    }
32. }
33.
```

MemoryController.cpp

```
1. void MemoryController::updateBankState()
2. {
3.     for (size_t i = 0; i < config.NUM_RANKS; i++)
4.     {
5.         for (size_t j = 0; j < config.NUM_BANKS; j++)
6.         {
7.             if (bankStates[i][j].stateChangeCountdown > 0)
8.             {
9.                 // decrement counters
10.                bankStates[i][j].stateChangeCountdown--;
11.            }
12.            // if counter has reached 0, change state
13.            if (bankStates[i][j].stateChangeCountdown == 0)
14.            {
15.                switch (bankStates[i][j].lastCommand)
16.                {
17.                    case REF:
18.                    case RFCSB:
19.                    case PRECHARGE:
20.                        bankStates[i][j].currentBankState = Idle;
21.                        break;
22.                    default:
23.                        break;
24.                }
25.            }
26.        }
27.    }
28. }
29.
```

MemoryController.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → MultiChannelMemorySystem::actual_update() → MemorySystem::update() → MemoryController::update()

2. Command Bus

- DRAM의 Command Bus에 실린 packet이 물리적으로 Rank에 도달하는데 걸리는 cycle을 count
- outgoingCmdPacket: 현재 Command Bus에 있는 command packet → cmdCyclesLeft: bus transfer latency(ex. tCMD cycle)을 추적
- cmdCyclesLeft==0 시, Rank에 도착한 것이기에 ranks[...]→receiveFromBus()를 호출 → outgoingCmdPacket = NULL로 초기화

```
1. void MemoryController::update()
2. {
3.     updateBankState();
4.     // check for outgoing command packets and handle countdowns
5.     if (outgoingCmdPacket != NULL)
6.     {
7.         cmdCyclesLeft--;
8.         if (cmdCyclesLeft == 0) // packet is ready to be received by rank
9.         {
10.            (*ranks)[outgoingCmdPacket->rank]->receiveFromBus(outgoingCmdPacket);
11.            outgoingCmdPacket = NULL;
12.        }
13.    }
14.
15.    // check for outgoing data packets and handle countdowns
16.    if (outgoingDataPacket != NULL)
17.    {
18.        dataCyclesLeft--;
19.        if (dataCyclesLeft == 0)
20.        {
21.            // inform upper levels that a write is done
22.            if (parentMemorySystem->WriteDataDone != NULL)
23.            {
24.                (*parentMemorySystem->WriteDataDone)(parentMemorySystem->systemID,
25.                outgoingDataPacket->physicalAddress,
26.                currentClockCycle);
27.            }
28.            parentMemorySystem->numOnTheFlyTransactions--;
29.            (*ranks)[outgoingDataPacket->rank]->receiveFromBus(outgoingDataPacket);
30.            outgoingDataPacket = NULL;
31.        }
32.    }
33. }
```

MemoryController.cpp

```
1. void Rank::receiveFromBus(BusPacket* packet)
2. {
3.     if (DEBUG_BUS)
4.     {
5.         PRINTN(" -- R" << getChanId() << " Receiving On Bus   : ");
6.         packet->print();
7.     }
8.     if (VERIFICATION_OUTPUT)
9.     {
10.        packet->print(currentClockCycle, false);
11.    }
12.    if (!pimRank->isReservedRA(packet->row))
13.    {
14.        check(packet);
15.        updateState(packet);
16.    }
17.    sendToBank(packet);
18. }
```

Rank.cpp

[cf] packet 구성

```
cmdPacket
패킷 구성:
busPacketType: ACTIVATE
physicalAddress: 0x12345600
rank: 0
bank: 2
row: 1024
column: (사용 안 함)
data: NULL

WRITE 를 위한 DataPacket
패킷 구성:
busPacketType: DATA
physicalAddress: 0x12345600
rank: 0
bank: 2
row: 1024
column: 8
data: [0xAF, 0xBE, 0xAD, ...] (64 바이트 분량의 실제 값) or NULL
```

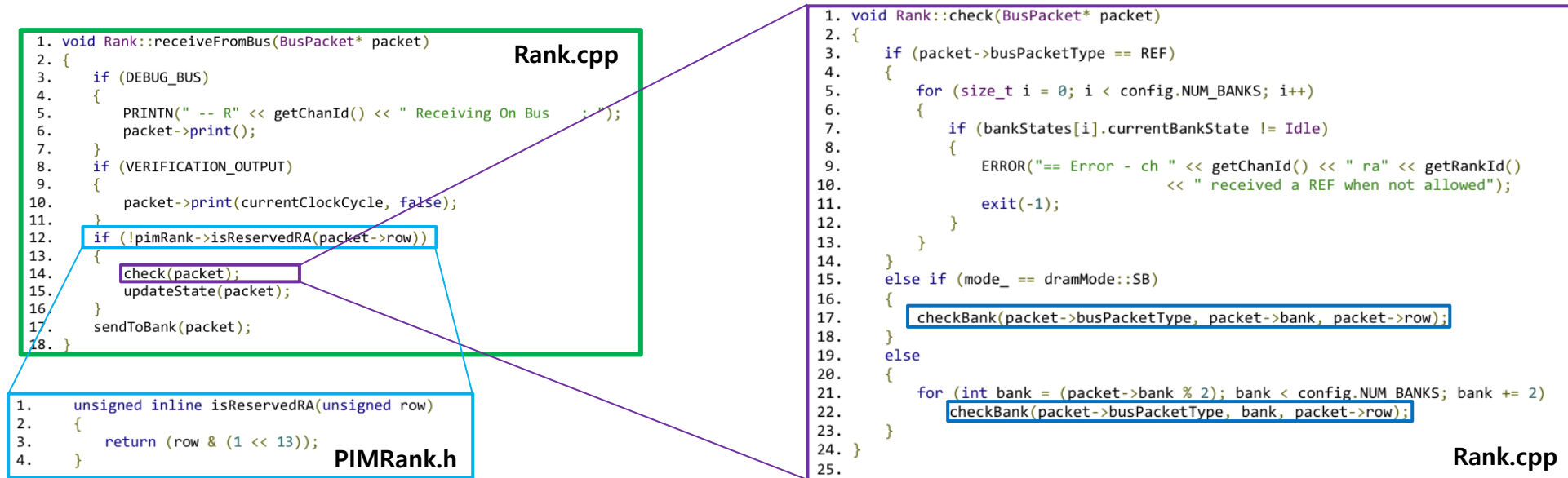
Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → ... → **MemoryController::update()** → **Rank::receiveFromBus()**

2. Command Bus → receiveFromBus()

- 명령이 PIM 연산을 위한 Reserved Row Address를 target하는지 확인 → 일반 DRAM 동작이 아닌 특수 mode라면 표준 timing check를 건너뛴
- 현재 Bank State에서 이 명령이 실행 가능한지 검사 → 다음 page에 계속



Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → ... → **MemoryController::update()** → **Rank::receiveFromBus()**
 - 현재 Bank State에서 이 명령이 실행 가능한지 검사
 - REF: 모든 Bank가 Idle 상태(open row가 없는 상태) → RowActive 상태인 Bank가 있다면 REFRESH 허용 X
 - Single Bank mode(= 일반적인 DRAM 동작) 검사 → packet에 명시된 단 1개의 Bank에 대해서만 timing을 check
 - HAB/PIM Mode 검사 (= Multi Bank Mode) → 여러 Bank를 동시에 제어 (모든 짝수 Bank 또는 모든 홀수 Bank를 순회)

```
1. void Rank::check(BusPacket* packet)
2. {
3.     if (packet->busPacketType == REF)
4.     {
5.         for (size_t i = 0; i < config.NUM_BANKS; i++)
6.         {
7.             if (bankStates[i].currentBankState != Idle)
8.             {
9.                 ERROR("== Error - ch " << getChanId() << " ra" << getRankId()
10.                    << " received a REF when not allowed");
11.                 exit(-1);
12.             }
13.         }
14.     }
15.     else if (mode_ == dramMode::SB)
16.     {
17.         checkBank(packet->busPacketType, packet->bank, packet->row);
18.     }
19.     else
20.     {
21.         for (int bank = (packet->bank % 2); bank < config.NUM_BANKS; bank += 2)
22.             checkBank(packet->busPacketType, bank, packet->row);
23.     }
24. }
25.
```

Rank.cpp

```
1. void Rank::checkBank(BusPacketType type, int bank, int row)
2. {
3.     switch (type)
4.     {
5.         case READ:
6.             if (bankStates[bank].currentBankState != RowActive ||
7.                 currentClockCycle < bankStates[bank].nextRead ||
8.                 row != bankStates[bank].openRowAddress)
9.             {
10.                 ERROR("== Error - ch " << getChanId() << " ra" << getRankId() << " ba" << bank
11.                    << " received a READ when not allowed @ "
12.                    << currentClockCycle);
13.                 exit(-1);
14.             }
15.             break;
16.         case WRITE:
17.             if (bankStates[bank].currentBankState != RowActive ||
18.                 currentClockCycle < bankStates[bank].nextWrite ||
19.                 row != bankStates[bank].openRowAddress)
20.             {
21.                 ERROR("== Error - ch " << getChanId() << " ra" << getRankId() << " ba" << bank
22.                    << " received a WRITE when not allowed @ "
23.                    << currentClockCycle);
24.                 bankStates[bank].print();
25.                 exit(-1);
26.             }
27.             break;
28.     }
29. }
```

```
29. case ACTIVATE:
30.     if (bankStates[bank].currentBankState != Idle ||
31.         currentClockCycle < bankStates[bank].nextActivate)
32.     {
33.         ERROR("== Error - ch " << getChanId() << " ra" << getRankId() << " ba" << bank
34.            << " received a ACT when not allowed @ "
35.            << currentClockCycle);
36.         bankStates[bank].print();
37.         exit(-1);
38.     }
39.     break;
40.
41. case PRECHARGE:
42.     if (bankStates[bank].currentBankState != RowActive ||
43.         currentClockCycle < bankStates[bank].nextPrecharge)
44.     {
45.         ERROR("== Error - ch " << getChanId() << " ra" << getRankId() << " ba" << bank
46.            << " received a PRE when not allowed @ "
47.            << currentClockCycle);
48.         exit(-1);
49.     }
50.     break;
51. case DATA:
52.     break;
53. default:
54.     ERROR("== Error - Unknown BusPacketType trying to be sent to Bank");
55.     exit(-1);
56.     break;
57. }
58. }
```

Rank.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → ... → **MemoryController::update()** → **Rank::receiveFromBus()** → **Rank::updateState()**

2. Command Bus → receiveFromBus()

- updateState()** : command을 받을 수 있으므로, 다음 command가 언제 들어올 수 있는지 timing을 update (현재 command는 실행될 예정 → timing update 필요)

- updateBank()**

1. REFRESH 시, Refresh 중에는 ACTIVATE command를 issue 못하기에, ACTIVATE command를 언제부터 issue가능한지 timing update

- 다음 page에 계속

```
1. void Rank::receiveFromBus(BusPacket* packet)
2. {
3.     if (DEBUG_BUS)
4.     {
5.         PRINTN(" -- R" << getChanId() << " Receiving On Bus : ");
6.         packet->print();
7.     }
8.     if (VERIFICATION_OUTPUT)
9.     {
10.        packet->print(currentClockCycle, false);
11.    }
12.    if (!pimRank->isReservedRA(packet->row))
13.    {
14.        check(packet);
15.        updateState(packet);
16.    }
17.    sendToBank(packet);
18. }
```

Rank.cpp

```
1. void Rank::updateState(BusPacket* packet)
2. {
3.     auto addrMapping = config.addrMapping;
4.     if (packet->busPacketType == REF)
5.     {
6.         refreshWaiting = false;
7.         for (size_t i = 0; i < config.NUM_BANKS; i++)
8.         {
9.             bankStates[i].nextActivate = currentClockCycle + config.tRFC;
10.        }
11.    }
12.    else if (mode_ == dramMode::SB)
13.    {
14.        for (int bank = 0; bank < config.NUM_BANKS; bank++)
15.        {
16.            updateBank(packet->busPacketType, bank, packet->row, bank == packet->bank,
17.                addrMapping.isSameBankgroup(bank, packet->bank));
18.        }
19.    }
20.    else
21.    {
22.        for (int bank = 0; bank < config.NUM_BANKS; bank++)
23.        {
24.            updateBank(packet->busPacketType, bank, packet->row, (bank % 2) == packet->bank, true);
25.        }
26.    }
27. }
```

Rank.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

• measureCycle() → ... → **MemoryController::update()** → **Rank::receiveFromBus()** → **Rank::updateState()** → **Rank::updateBank()**

• **updateBank()**

2. Single Bank mode(= 일반적인 DRAM 동작) 검사 → 모든 Bank에 대해 updateBank를 호출하되, 내 명령의 직접적인 대상인지 정보를 넘김

- 대상 Bank: 명령에 따른 직접적인 latency(ex. tRP, tRC) 적용
- 같은 Bank Group: Bank Group 간 timing(ex. tRRDL, tCCDL) 적용
- 다른 Bank Group: 최소한의 간섭 timing(ex. tRRDS, tCCDS) 적용
- 다음 page에 계속

```
1. void Rank::updateState(BusPacket* packet)
2. {
3.     auto addrMapping = config.addrMapping;
4.     if (packet->busPacketType == REF)
5.     {
6.         refreshWaiting = false;
7.         for (size_t i = 0; i < config.NUM_BANKS; i++)
8.         {
9.             bankStates[i].nextActivate = currentClockCycle + config.tRFC;
10.        }
11.    }
12.    else if (mode_ == dramMode::SB)
13.    {
14.        for (int bank = 0; bank < config.NUM_BANKS; bank++)
15.        {
16.            updateBank(packet->busPacketType, bank, packet->row, bank == packet->bank,
17.                       addrMapping.isSameBankgroup(bank, packet->bank));
18.        }
19.    }
20.    else
21.    {
22.        for (int bank = 0; bank < config.NUM_BANKS; bank++)
23.        {
24.            updateBank(packet->busPacketType, bank, packet->row, (bank % 2) == packet->bank, true);
25.        }
26.    }
27. }
```

Rank.cpp

```
1. void Rank::updateBank(BusPacketType type, int bank, int row, bool targetBank, bool targetBankgroup)
2. {
3.     switch (type)
4.     {
5.         case READ:
6.             if (targetBank)
7.             {
8.                 bankStates[bank].nextPrecharge = max(bankStates[bank].nextPrecharge,
9.                                                         currentClockCycle + config.READ_TO_PRE_DELAY);
10.            }
11.            if (targetBankgroup)
12.            {
13.                bankStates[bank].nextRead =
14.                    max(bankStates[bank].nextRead,
15.                        currentClockCycle + max(config.tCCDL, config.BL / 2));
16.            }
17.            else
18.            {
19.                bankStates[bank].nextRead =
20.                    max(bankStates[bank].nextRead,
21.                        currentClockCycle + max(config.tCCDS, config.BL / 2));
22.            }
23.            bankStates[bank].nextWrite =
24.                max(bankStates[bank].nextWrite, currentClockCycle + config.READ_TO_WRITE_DELAY);
25.            break;
26.        case WRITE:
27.            // update state table
28.            if (targetBank)
29.            {
30.                bankStates[bank].nextPrecharge = max(bankStates[bank].nextPrecharge,
31.                                                         currentClockCycle + config.WRITE_TO_PRE_DELAY);
32.            }
33.            if (targetBankgroup)
34.            {
35.                bankStates[bank].nextRead =
36.                    max(bankStates[bank].nextRead,
37.                        currentClockCycle + config.WRITE_TO_READ_DELAY_B_LONG);
38.                bankStates[bank].nextWrite =
39.                    max(bankStates[bank].nextWrite,
40.                        currentClockCycle + max(config.BL / 2, config.tCCDL));
41.            }
42.            else
43.            {
44.                bankStates[bank].nextRead =
45.                    max(bankStates[bank].nextRead,
46.                        currentClockCycle + config.WRITE_TO_READ_DELAY_B_SHORT);
47.                bankStates[bank].nextWrite =
48.                    max(bankStates[bank].nextWrite,
49.                        currentClockCycle + max(config.BL / 2, config.tCCDS));
50.            }
51.            break;
52.        case ACTIVATE:
53.            if (targetBank)
54.            {
55.                bankStates[bank].currentBankState = RowActive;
56.                bankStates[bank].nextActivate = currentClockCycle + config.tRC;
57.                bankStates[bank].openRowAddress = row;
58.                bankStates[bank].nextWrite = currentClockCycle + (config.tRCdWR - config.AL);
59.                bankStates[bank].nextRead = currentClockCycle + (config.tRCdRD - config.AL);
60.                bankStates[bank].nextPrecharge = currentClockCycle + config.tRAS;
61.            }
62.            else
63.            {
64.                bankStates[bank].nextActivate =
65.                    (targetBankgroup
66.                     ? max(bankStates[bank].nextActivate, currentClockCycle + config.tRRDL)
67.                     : max(bankStates[bank].nextActivate, currentClockCycle + config.tRRDS));
68.                break;
69.        case PRECHARGE:
70.            if (targetBank)
71.            {
72.                bankStates[bank].currentBankState = Idle;
73.                bankStates[bank].nextActivate =
74.                    max(bankStates[bank].nextActivate, currentClockCycle + config.tRP);
75.            }
76.            break;
77.        case DATA:
78.            break;
79.        default:
80.            ERROR("Error - Unknown BusPacketType trying to be sent to Bank");
81.            exit(0);
82.            break;
83.    }
84. }
```

Rank.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

• measureCycle() → ... → **MemoryController::update()** → **Rank::receiveFromBus()** → **Rank::updateState()** → **Rank::updateBank()**

• **updateBank()**

3. HAB/PIM mode (= Multi Bank Mode) → 짝수 또는 홀수 번호의 Bank들을 동시에 Target Bank로 취급하여 업데이트

- 대상 Bank: 명령에 따른 직접적인 latency(ex. tRP, tRC) 적용
- 같은 Bank Group: Bank Group 간 timing(ex. tRRDL, tCCDL) 적용
- 다른 Bank Group: 최소한의 간섭 timing(ex. tRRDS, tCCDS) 적용

```
1. void Rank::updateState(BusPacket* packet)
2. {
3.     auto addrMapping = config.addrMapping;
4.     if (packet->busPacketType == REF)
5.     {
6.         refreshWaiting = false;
7.         for (size_t i = 0; i < config.NUM_BANKS; i++)
8.         {
9.             bankStates[i].nextActivate = currentClockCycle + config.tRFC;
10.        }
11.    }
12.    else if (mode_ == dramMode::SB)
13.    {
14.        for (int bank = 0; bank < config.NUM_BANKS; bank++)
15.        {
16.            updateBank(packet->busPacketType, bank, packet->row, bank == packet->bank,
17.                       addrMapping.isSameBankgroup(bank, packet->bank));
18.        }
19.    }
20.    else
21.    {
22.        for (int bank = 0; bank < config.NUM_BANKS; bank++)
23.        {
24.            updateBank(packet->busPacketType, bank, packet->row, (bank % 2) == packet->bank, true);
25.        }
26.    }
27. }
```

Rank.cpp

```
1. void Rank::updateBank(BusPacketType type, int bank, int row, bool targetBank, bool targetBankgroup)
2. {
3.     switch (type)
4.     {
5.         case READ:
6.             if (targetBank)
7.             {
8.                 bankStates[bank].nextPrecharge = max(bankStates[bank].nextPrecharge,
9.                                                         currentClockCycle + config.READ_TO_PRE_DELAY);
10.            }
11.            if (targetBankgroup)
12.            {
13.                bankStates[bank].nextRead =
14.                    max(bankStates[bank].nextRead,
15.                        currentClockCycle + max(config.tCCDL, config.BL / 2));
16.            }
17.            else
18.            {
19.                bankStates[bank].nextRead =
20.                    max(bankStates[bank].nextRead,
21.                        currentClockCycle + max(config.tCCDS, config.BL / 2));
22.            }
23.            bankStates[bank].nextWrite =
24.                max(bankStates[bank].nextWrite, currentClockCycle + config.READ_TO_WRITE_DELAY);
25.            break;
26.        case WRITE:
27.            // update state table
28.            if (targetBank)
29.            {
30.                bankStates[bank].nextPrecharge = max(bankStates[bank].nextPrecharge,
31.                                                         currentClockCycle + config.WRITE_TO_PRE_DELAY);
32.            }
33.            if (targetBankgroup)
34.            {
35.                bankStates[bank].nextRead =
36.                    max(bankStates[bank].nextRead,
37.                        currentClockCycle + config.WRITE_TO_READ_DELAY_B_LONG);
38.                bankStates[bank].nextWrite =
39.                    max(bankStates[bank].nextWrite,
40.                        currentClockCycle + max(config.BL / 2, config.tCCDL));
41.            }
42.            else
43.            {
44.                bankStates[bank].nextRead =
45.                    max(bankStates[bank].nextRead,
46.                        currentClockCycle + config.WRITE_TO_READ_DELAY_B_SHORT);
47.                bankStates[bank].nextWrite =
48.                    max(bankStates[bank].nextWrite,
49.                        currentClockCycle + max(config.BL / 2, config.tCCDS));
50.            }
51.            break;
52.        case ACTIVATE:
53.            if (targetBank)
54.            {
55.                bankStates[bank].currentBankState = RowActive;
56.                bankStates[bank].nextActivate = currentClockCycle + config.tRC;
57.                bankStates[bank].openRowAddress = row;
58.                bankStates[bank].nextWrite = currentClockCycle + (config.tRCdWR - config.AL);
59.                bankStates[bank].nextRead = currentClockCycle + (config.tRCdRD - config.AL);
60.                bankStates[bank].nextPrecharge = currentClockCycle + config.tRAS;
61.            }
62.            else
63.            {
64.                bankStates[bank].nextActivate =
65.                    (targetBankgroup
66.                     ? max(bankStates[bank].nextActivate, currentClockCycle + config.tRRDL)
67.                     : max(bankStates[bank].nextActivate, currentClockCycle + config.tRRDS));
68.                break;
69.            }
70.        case PRECHARGE:
71.            if (targetBank)
72.            {
73.                bankStates[bank].currentBankState = Idle;
74.                bankStates[bank].nextActivate =
75.                    max(bankStates[bank].nextActivate, currentClockCycle + config.tRP);
76.            }
77.            break;
78.        case DATA:
79.            break;
80.        default:
81.            ERROR("Error - Unknown BusPacketType trying to be sent to Bank");
82.            exit(0);
83.            break;
84.    }
85. }
```

Rank.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → ... → MemoryController::update() → Rank::receiveFromBus() → Rank::sendToBank()

2. Command Bus → receiveFromBus()

- sendToBank(): Command가 실제 Bank 또는 PIM에서 실행
- 다음 page에서 sendToBank() 분석

```
1. void Rank::receiveFromBus(BusPacket* packet)
2. {
3.     if (DEBUG_BUS)
4.     {
5.         PRINTN(" -- R" << getChanId() << " Receiving On Bus   ");
6.         packet->print();
7.     }
8.     if (VERIFICATION_OUTPUT)
9.     {
10.        packet->print(currentClockCycle, false);
11.    }
12.    if (!pimRank->isReservedRA(packet->row))
13.    {
14.        check(packet);
15.        updateState(packet);
16.    }
17.    sendToBank(packet);
18. }
```

Rank.cpp

```
1. void Rank::sendToBank(BusPacket* packet)
2. {
3.     switch (packet->busPacketType)
4.     {
5.         case READ:
6.             if (mode_ == dramMode::SB)
7.                 readSb(packet);
8.             else if (mode_ == dramMode::HAB_PIM && pimRank->isToggleCond(packet))
9.                 pimRank->doPIM(packet);
10.            else
11.            {
12.                pimRank->readHab(packet);
13.                packet->busPacketType = DATA;
14.                readReturnPacket.push_back(packet);
15.                readReturnCountdown.push_back(config.RL);
16.                break;
17.            }
18.        case WRITE:
19.            if (mode_ == dramMode::SB)
20.                writeSb(packet);
21.            else if (mode_ == dramMode::HAB_PIM && pimRank->isToggleCond(packet))
22.                pimRank->doPIM(packet);
23.            else
24.            {
25.                pimRank->writeHab(packet);
26.                delete (packet);
27.                break;
28.            }
29.        case ACTIVATE:
30.            if (DEBUG_CMD_TRACE)
31.            {
32.                PRINTC(getModeColor(), OUTLOG_ALL("ACTIVATE") << " tag : " << packet->tag);
33.            }
34.            if (mode_ == dramMode::SB && packet->row == config.PIM_ABMR_RA &&
35.                packet->column == 0x1f)
36.            {
37.                abmr1Even_ = (packet->bank == 0) ? true : abmr1Even_;
38.                abmr1Odd_ = (packet->bank == 1) ? true : abmr1Odd_;
39.                abmr2Even_ = (packet->bank == 0) ? true : abmr2Even_;
40.                abmr2Odd_ = (packet->bank == 1) ? true : abmr2Odd_;
41.            }
42.            if ((config.NUM_BANKS <= 2 && abmr1Even_ && abmr1Odd_) ||
43.                (config.NUM_BANKS > 2 && abmr1Even_ && abmr1Odd_ && abmr2Even_ && abmr2Odd_))
44.            {
45.                abmr1Even_ = abmr1Odd_ = abmr2Even_ = abmr2Odd_ = false;
46.                mode_ = dramMode::HAB;
47.                if (DEBUG_CMD_TRACE)
48.                {
49.                    PRINTC(RED, OUTLOG_CH_RA("HAB") << " tag : " << packet->tag);
50.                }
51.            }
52.            delete (packet);
53.            break;
54.        case PRECHARGE:
55.            if (DEBUG_CMD_TRACE)
56.            {
57.                if (mode_ == dramMode::SB || packet->bank < 2)
58.                {
59.                    PRINTC(getModeColor(), OUTLOG_PRECHARGE("PRECHARGE"));
60.                }
61.                else
62.                {
63.                    PRINTC(GRAY, OUTLOG_PRECHARGE("PRECHARGE"));
64.                }
65.            }
66.            if (mode_ == dramMode::HAB && packet->row == config.PIM_SBMR_RA)
67.            {
68.                sbmr1_ = (packet->bank == 0) ? true : sbmr1_;
69.                sbmr2_ = (packet->bank == 1) ? true : sbmr2_;
70.                if (sbmr1_ && sbmr2_)
71.                {
72.                    sbmr1_ = sbmr2_ = false;
73.                    mode_ = dramMode::SB;
74.                    if (DEBUG_CMD_TRACE)
75.                    {
76.                        PRINTC(RED, OUTLOG_CH_RA("SB mode"));
77.                    }
78.                }
79.            }
80.            delete (packet);
81.            break;
82.        case REF:
83.            refreshWaiting = false;
84.            if (DEBUG_CMD_TRACE)
85.            {
86.                PRINTC(OUTLOG_CH_RA("REF"));
87.            }
88.            delete (packet);
89.            break;
90.        case DATA:
91.            delete (packet);
92.            break;
93.        default:
94.            ERROR("Error - Unknown BusPacketType trying to be sent to Bank");
95.            exit(0);
96.            break;
97.    }
98. }
```

Rank.cpp

Samsung PIM Simulator hbm read test flow

□ hbm read test flow – measureCycle()

- measureCycle() → MultiChannelMemorySystem::actual_update() → MemorySystem::update() → MemoryController::update()

3. Write Data Bus

- Host(MC)에서 DRAM(Rank)으로 가는 Write Data 흐름을 관리
- dataCyclesLeft: data Burst가 유지되는 시간을 추적 → WriteDataDone: counter가 0이 되면, MC는 bus transfer가 끝났음을 MemorySystem에 알림
- Transaction 관리: numOnTheFlyTransactions를 줄여서, 현재 처리 중인 작업 하나가 끝났음을 기록

```
1. void MemoryController::update()
2. {
3.     updateBankState();
4.     // check for outgoing command packets and handle countdowns
5.     if (outgoingCmdPacket != NULL)
6.     {
7.         cmdCyclesLeft--;
8.         if (cmdCyclesLeft == 0) // packet is ready to be received by rank
9.         {
10.            (*ranks)[outgoingCmdPacket->rank]->receiveFromBus(outgoingCmdPacket);
11.            outgoingCmdPacket = NULL;
12.        }
13.    }
14.
15.    // check for outgoing data packets and handle countdowns
16.    if (outgoingDataPacket != NULL)
17.    {
18.        dataCyclesLeft--;
19.        if (dataCyclesLeft == 0)
20.        {
21.            // inform upper levels that a write is done
22.            if (parentMemorySystem->WriteDataDone != NULL)
23.            {
24.                (*parentMemorySystem->WriteDataDone)(parentMemorySystem->systemID,
25.                outgoingDataPacket->physicalAddress,
26.                currentClockCycle);
27.            }
28.            parentMemorySystem->numOnTheFlyTransactions--;
29.            (*ranks)[outgoingDataPacket->rank]->receiveFromBus(outgoingDataPacket);
30.            outgoingDataPacket = NULL;
31.        }
32.    }
33. }
```

[cf] packet 구성

```
cmdPacket
패킷 구성:
  busPacketType: ACTIVATE
  physicalAddress: 0x12345600
  rank: 0
  bank: 2
  row: 1024
  column: (사용 안 함)
  data: NULL

WRITE 를 위한 DataPacket
패킷 구성:
  busPacketType: DATA
  physicalAddress: 0x12345600
  rank: 0
  bank: 2
  row: 1024
  column: 8
  data: [0xAF, 0xBE, 0xAD, ...] (64 바이트 분량의 실제 값) or NULL
```

Samsung PIM Simulator gemv test flow

□ gemv test flow

- Samsung PIM Simulator는 GoogleTest(gtest)기반의 simulation test를 수행함
- 따라서, 이 flow를 Samsung PIM Simulator가 gemv operation을 어떻게 test하는지 따라가며 알아본다

./sim --gtest_filter=PIMKernelFixture.gemv

```
1. #include <bitset>
2. #include <cstdlib>
3. #include <iostream>
4. #include <limits>
5. #include <memory>
6. #include <random>
7.
8. #include "Burst.h"
9. #include "MultiChannelMemorySystem.h"
10. #include "gtest/gtest.h"
11. #include "tests/PIMKernel.h"
12. #include "tests/TestCases.h"
13.
14. using namespace std;
15.
16. int main(int argc, char* argv[])
17. {
18.     if (argc > 1)
19.     {
20.         ::testing::InitGoogleTest(&argc, argv);
21.     }
22.     else
23.     {
24.         ::testing::InitGoogleTest(&argc, argv);
25.         testing::GTEST_FLAG(filter) = "-*bw*.*";
26.     }
27.
28.     return RUN_ALL_TESTS();
29. }
```

MainTest.cpp

```
1. TEST_F(PIMKernelFixture, gemv)
2. {
3.     shared_ptr<PIMKernel> kernel = make_pim_kernel();
4.
5.     uint32_t batch_size = 1;
6.     uint32_t output_dim = 4096;
7.     uint32_t input_dim = 1024;
8.
9.     DataDim *dim_data = new DataDim(KernelType::GEMV, batch_size, output_dim, input_dim, true);
10.    dim_data->printDim(KernelType::GEMV);
11.
12.    reduced_result_ = new BurstType[dim_data->dimTobShape(output_dim)];
13.    result_ = getResultPIM(KernelType::GEMV, dim_data, kernel, result_);
14.
15.    testStatsClear();
16.    expectAccuracy(KernelType::GEMV, output_dim, dim_data->output_npbst_,
17.                  dim_data->getNumElementsPerBlocks());
18.
19.    delete[] result_;
20.    delete[] reduced_result_;
21.    delete dim_data;
22. }
```

KernelTestCases.cpp

Samsung PIM Simulator gemv test flow

□ gemv test flow

- Samsung PIM Simulator는 GoogleTest(gtest)기반의 simulation test를 수행함
- 따라서, 이 flow를 Samsung PIM Simulator가 gemv operation을 어떻게 test하는지 따라가며 알아본다

```
1. TEST_F(PIMKernelFixture, gemv)
2. {
3.     shared_ptr<PIMKernel> kernel = make_pim_kernel();
4.
5.     uint32_t batch_size = 1;
6.     uint32_t output_dim = 4096;
7.     uint32_t input_dim = 1024;
8.
9.     DataDim *dim_data = new DataDim(KernelType::GEMV, batch_size, output_dim, input_dim, true);
10.    dim_data->printDim(KernelType::GEMV);
11.
12.    reduced_result_ = new BurstType[dim_data->dimTobShape(output_dim)];
13.    result_ = getResultPIM(KernelType::GEMV, dim_data, kernel, result_);
14.
15.    testStatsClear();
16.    expectAccuracy(KernelType::GEMV, output_dim, dim_data->output_npbst_,
17.                  dim_data->getNumElementsPerBlocks());
18.
19.    delete[] result_;
20.    delete[] reduced_result_;
21.    delete dim_data;
22. }
```

```
1. shared_ptr<PIMKernel> make_pim_kernel()
2. {
3.     shared_ptr<MultiChannelMemorySystem> mem = make_shared<MultiChannelMemorySystem>(
4.         "ini/HBM2_samsung_2M_16B_x64.ini", "system_hbm_64ch.ini", ".", "example_app",
5.         256 * 64 * 2);
6.     int numPIMChan = 64;
7.     int numPIMRank = 1;
8.     shared_ptr<PIMKernel> kernel = make_shared<PIMKernel>(mem, numPIMChan, numPIMRank);
9.
10.    return kernel;
11. }
```

KernelTestCases.cpp